

AGBELUSI AKINRINSOLA JOSHUA

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Research Summary

Seismology researcher with a physics background, applying machine learning to earthquake detection and phase picking. Experienced in building and validating automated earthquake catalogues from continuous seismic data, seismic waveform analysis, and local earthquake tomography workflows. Broader interests span computational and exploration geophysics and the geophysics of subsurface energy systems.

Research Interests

Machine learning for seismic event detection, phase picking, and classification; earthquake catalogue construction and validation; local earthquake tomography; computational and observational seismology; volcanic and geothermal seismicity.

Education

MSc, Petroleum Science and Engineering 2025 - 2027 (expected)

Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia

Relevant courses: Petroleum Geophysics, Machine Learning, Applied Geomechanics, Thermal Petrophysics and Geothermy, Reservoir Rock Characterisation, Reservoir Petrophysics and Geomodelling.

Graduate Programme in Geophysics (non-degree) 2023 - 2025

University of Lagos, Akoka, Nigeria

Coursework: Seismic Methods, Computational Geophysics, Borehole Geophysics, Potential Field Methods, Electromagnetic Methods.

BSc, Physics with Electronics 2016 - 2020

Crawford University, Igbesa, Nigeria

CGPA 4.22/5.0 (2nd of 15). Projects: ionospheric response during the geomagnetic storms of April 2010 and March 2015; wireless power transfer with a Tesla coil.

Research Experience

Graduate Researcher, Earthquake Seismology 2026 - Present

Skolkovo Institute of Science and Technology, Moscow, Russia

Funded research assistant on Russian Science Foundation project (grant #26-17-00179), "Identification of geothermal resources in volcanic areas using seismic tomography methods".

- Built an automated machine-learning pipeline (PhaseNet and EQTransformer via SeisBench) for P- and S-wave phase picking on continuous three-component data from a 65-station network in the Mutnovsky-Gorely volcanic and geothermal region, Kamchatka.
- Applied the pipeline to the continuous data, detecting and locating over 3,300 high-quality earthquakes through automated association and quality filtering, a twenty-fold increase over the previous manually picked catalogue that fundamentally improved the tomographic resolution.
- Validated the automated catalogue against an expert manual catalogue, and tested a volcanic-trained model that recovered more expert arrivals than the standard models (about 65% against 43% of P picks).
- The catalogue underpins Prof. Koulakov's local earthquake tomography of the complex, a coauthored paper submitted to Journal of Geophysical Research: Solid Earth.

Graduate Research Assistant, Geophysics 2023 - 2025

University of Lagos, Department of Geophysics, Nigeria

- Conducted gravity and magnetic surveys of subsurface structures; processed and interpreted data in Oasis Montaj; supported undergraduate laboratory teaching.

Intern, Junior Physicist June - September 2019

Nigerian National Petroleum Corporation (NNPC), Lagos, Nigeria

- Assisted with seismic data acquisition, equipment calibration, and gravity surveys during field operations.

Projects

Tomo Workbench, Seismic Processing and Tomography Software (Project Lead) 2026 - Present

- Building a graphical tool that runs the full workflow, event association and location, catalogue cleaning, and 3D velocity inversion from a starting 1D velocity model.

Seismophone, Seismic Data Sonification (Project Lead) 2026 - Present

- Building a tool that converts seismic waveforms into audio to explore audio-based machine learning for earthquake detection and classification.

Cowllar, Cattle Estrus Detection (Technical Lead) 2026

- Built a non-invasive wearable with on-device machine-learning inference, combining behaviour, rumination, audio, and temperature signals; designed the sensor hardware, firmware, data pipeline, model, and a web dashboard, reaching 90.4 percent four-class accuracy.

Technical Skills

Seismology: seismic waveform processing, phase picking, MiniSEED and DIMAS data, SeisBench (PhaseNet, EQTransformer), ObsPy; earthquake catalogue construction and validation; gravity, magnetic, potential-field, and electromagnetic methods.

Machine learning and data analysis: classification, regression, feature extraction, model evaluation, deep-learning methods; Jupyter.

Programming: Python (NumPy, Pandas, SciPy, scikit-learn, ObsPy, Matplotlib), MATLAB, JavaScript, HTML/CSS; sensor firmware and data logging.

Petroleum geoscience: reservoir rock characterisation, petrophysics, well-log interpretation, thermal petrophysics.

Software: Oasis Montaj, Petrel (introductory), Microsoft Office, Surfer.

Publications

- Agbelusi, A. J., and Koulakov, I. (2026). Seismicity of the Bolshe-Bannye geothermal field, Kamchatka, from automated detection: capability and limits of a small-aperture network. *Geophysical Technologies*, submitted.
- Koulakov, I., Agbelusi, A. J., Stupina, T., Abkadyrov, I., Khmarin, E., and Chebrov, D. V. (2026). Crustal magmatic and hydrothermal system of the Mutnovsky-Gorely volcanic complex (Kamchatka) imaged by local earthquake tomography. *Journal of Geophysical Research: Solid Earth*, submitted.
- Okiye, S. E., Emekwisia, C. C., Igwe, E. S., Omofaye, V. I., Agbelusi, A. J., Ajibode, H. J., Dada, A. O., and Agbahiwe, O. K. (2025). Comparative analysis of thermal insulation properties of bricks made from local and industrial by-products. *American Journal of Applied Sciences and Engineering*, 6(3), 11-16. DOI: 10.5281/zenodo.16229391.

Awards and Leadership

- Fully funded merit scholarship, Skolkovo Institute of Science and Technology (2025).
- Best Student, Department of Physics, Crawford University (2020).
- Dean's List, College of Natural and Applied Sciences, Crawford University (2017-2019).
- President, Department of Physics Student Association, Crawford University (2020); Academic Director, Student Representative Council (2019-2020).

Teaching

Mathematics and Physics Teacher, Bammy College, Ogun State, Nigeria (2020-2022). Taught senior secondary Mathematics and Physics; founded a STEM club.

Referees

Prof. Ivan Koulakov

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